EXPERIMENT – 2

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DESIGN A UI WHERE USERS RECALL VISUAL ELEMENTS (e.g..,icons). EVALUATE THE EFFECT OF CHUNKING ON USER MEMORY

FRAME 1: INSTRUCTION / START SCREEN

Chunking Analysis of the Instruction Page

This screen introduces the Visual Memory Test and prepares users for the task using chunking principles.

1. Clear Task Focus

The title “Visual Memory Test” is placed at the top in a highlighted label.

The instruction “Observe the icons carefully and remember it.” gives one simple, direct action.

The brain icon visually reinforces that this is a memory-related activity.

2. Minimal Information (Reduced Cognitive Load)

Only essential information is shown.

No long paragraphs — just one instruction sentence.

This prevents overload and helps users focus only on remembering icons.

3. Visual Hierarchy

Title → Instruction → Start Button (top to bottom flow).

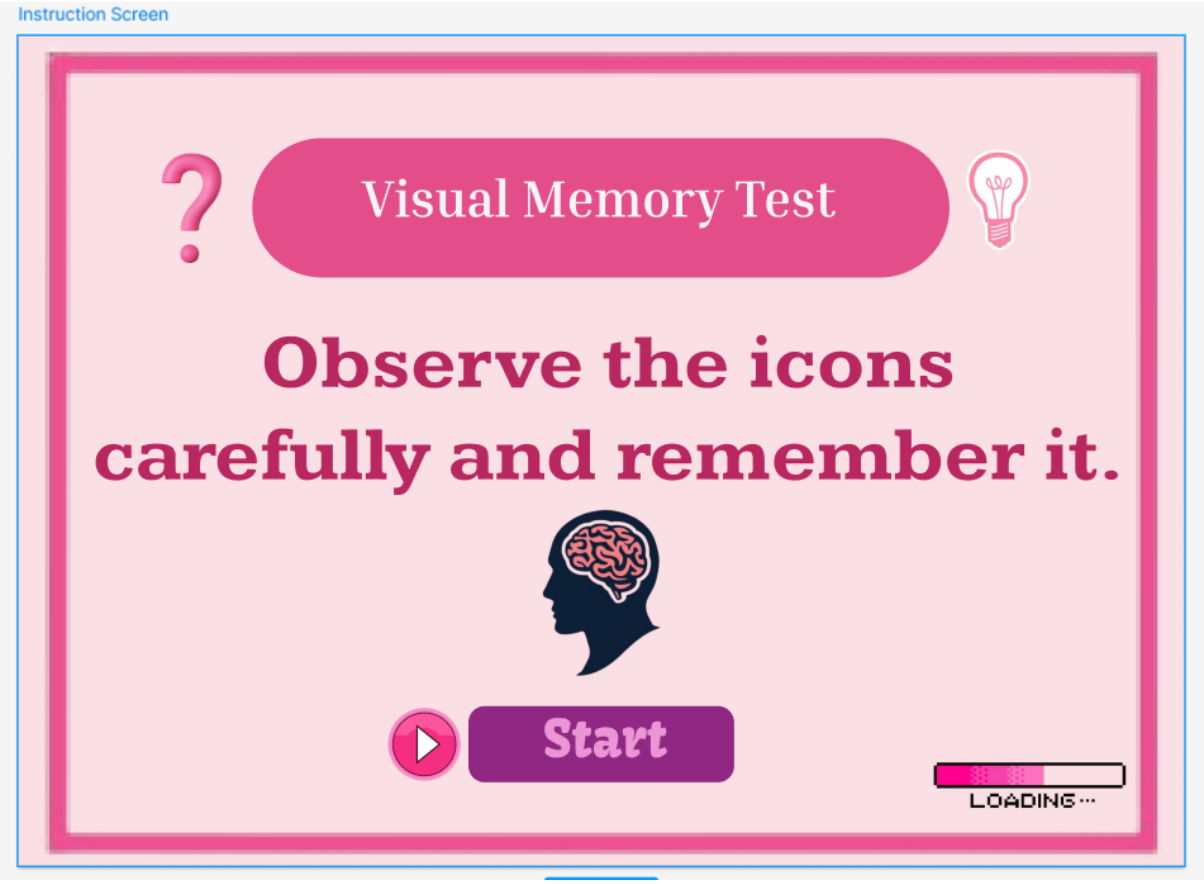
The “Start” button is large and centered, guiding the next step clearly.

4. Engaging Design

Soft pink theme keeps the interface friendly and non-stressful.

Simple graphics help users mentally prepare without distraction.

FRAME - 1 INSTRUCTION SCREEN/PAGE



FRAME 2 : CHUNKED ICONS

Chunking Analysis of the Observation Screen

This screen represents the main memory encoding stage where users observe a set of icons and try to remember them. The design of this screen supports the concept of chunking by helping users mentally organize the icons instead of viewing them as random symbols.

1. Purpose of the Screen

Users are given time to observe and memorize the displayed icons.

Instead of remembering each icon separately, the layout encourages users to group similar icons together mentally.

2. Use of Chunking Through Visual Grouping

The icons are arranged in separate sections, which naturally encourages the brain to treat each section as one group.

When information is grouped, users remember the category or theme first, and then the items inside it.

This reduces memory load because users recall groups of items instead of many individual icons.

3. Visual Structure Supports Memory

Boxes, spacing, and headings visually separate different sets of icons.

This separation acts as a mental boundary, making it easier for users to organize what they see.

The clean layout avoids clutter, so the brain can focus on patterns rather than confusion.

4. Attention and Focus

Since this is the only task on the screen, users focus completely on observing.

The “Next” button appears after viewing, which ensures users complete the observation phase before moving forward.

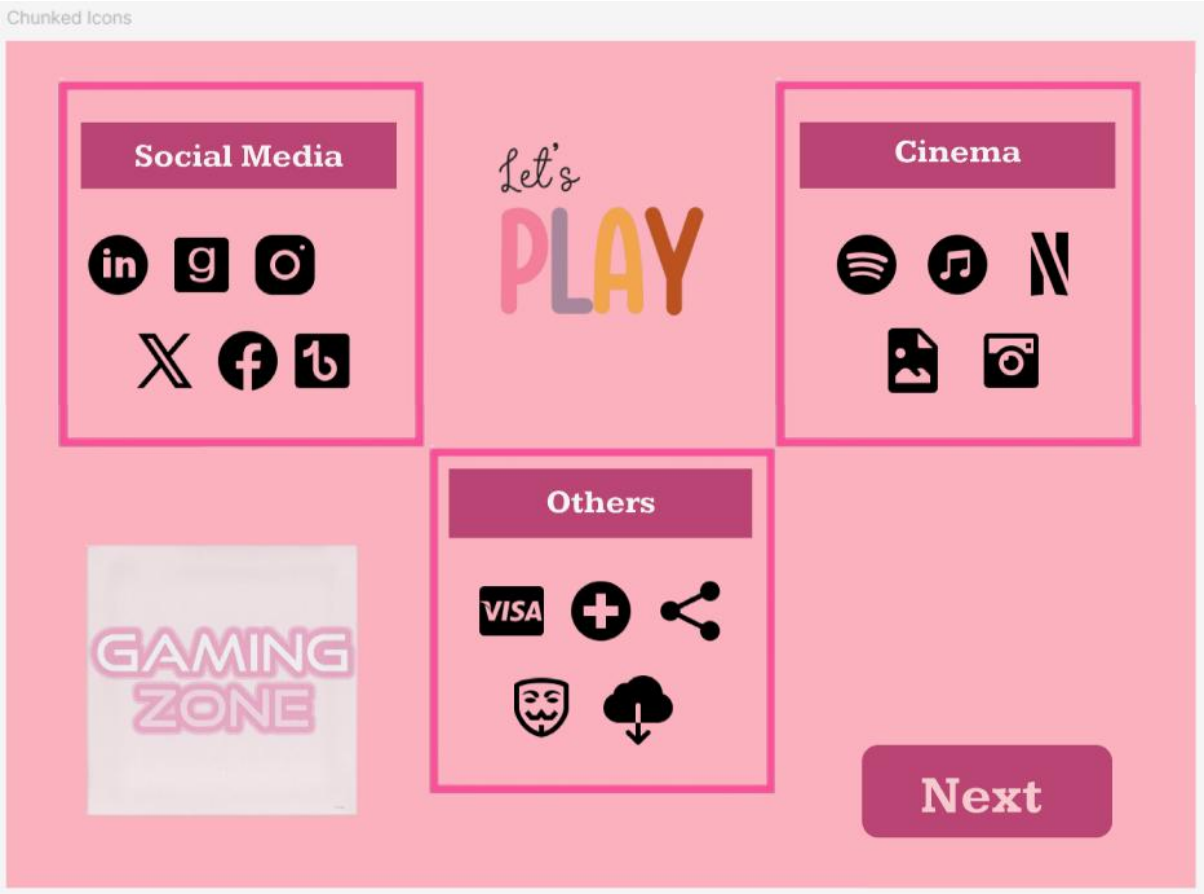
5. Cognitive Benefit of This Screen

Helps users create mental chunks while observing

Reduces cognitive overload

Improves short-term memory retention

FRAME 2 : CHUNKED ICONS



FRAME 3 : FIRST RECALL SCREEN

This screen is the first memory recall stage where users try to remember the icons they saw before.

1.Purpose of the Screen

Users must write the icons they remember from the previous screen. No icons are shown here, so this checks memory without any hints.

2. How Chunking Helps in Recall

Since the icons were grouped earlier, users may remember them in groups instead of one by one.

For example, a user might first remember a group like social media or cinema, and then recall the icons inside that group. This shows that chunking makes remembering easier.

3. Simple and Clear Design

A big blank box is given for users to type their answers.

There are no extra pictures or symbols, so users can fully focus on remembering.

The layout is neat and easy to understand.

4. Navigation

A Submit button is placed below the answer box.

Users press it after they finish recalling.

5. Memory Benefits

Helps test how well users remember

Shows if grouping helped memory

Measures the success of chunking

FRAME 3 : FIRST RECALL SCREEN

FRAME 4 : RECOGNITION SCREEN

Chunking Analysis of the Recognition Screen

This screen is the recognition stage where users select the icons they remember seeing earlier.

1. Purpose of the Screen

Users choose the icons they remember. Some icons are correct, and some are new distractors.

This tests how accurately users remember.

2. How Chunking Helps

Even though the icons are in a random order, users may remember them in groups from the earlier screen.

Remembering categories helps them identify the correct icons.

3. Layout

Icons are placed together in one area but not grouped, making users depend only on memory.

The clean design helps users focus while selecting.

4. Memory Benefit

Tests recognition memory

Measures how chunking helped recall even without visual grouping

FRAME 4 : RANDOM ICONS

FRAME 5: FINAL RECALL SCREEN

Chunking Analysis of the Final Recall Screen

This screen is the final memory recall stage where users again write the icons they remember.

<u>1. Purpose of the Screen</u>

Users must recall and write the icons from memory.

No options or hints are given.

This checks how much information stayed in memory.

2. Role of Chunking

Users are more likely to remember groups of icons instead of all icons one by one.

This shows how chunking helps in keeping information in memory for a longer time.

3. Simple Layout

A large blank area is given to write answers.

No extra pictures or symbols are shown, so users can focus fully.

4. Navigation

A Submit button is provided to finish the task.

5. Memory Benefit

Tests final memory strength

Shows which groups were remembered best

Measures the overall effect of chunking

FRAME 5 : FINAL RECALL SCREEN